Hat Matrix	**Pseudoinverse**
Note: For solving SVD with Hermitian Dilation	**SLR by QR decomposition**
Hermitian Dilation	**Group**
Ring	**Commutative Ring**
Ring with Identity	**Normal Equations**

$$\boldsymbol{A}^{+} = \left(\boldsymbol{A}^{*}\boldsymbol{A}\right)^{-1}\boldsymbol{A}^{*}$$

$$\boldsymbol{P} = \boldsymbol{X}\left(\boldsymbol{X}^{*}\boldsymbol{X}\right)^{-1}\boldsymbol{X}^{*} \in \mathbb{R}^{n\times n}$$

- orthogonal projector
- idempotent and symmetric

1. Decompose $\boldsymbol{X} = \hat{\boldsymbol{Q}}\hat{\boldsymbol{R}}$
2. Compute $\boldsymbol{t} = \hat{\boldsymbol{Q}}^{*}\boldsymbol{y}$
3. Use back substitution to solve $\hat{\boldsymbol{R}}\hat{\boldsymbol{\beta}} = \boldsymbol{t}$ for $\hat{\boldsymbol{\beta}}$

Look at HW10 #4

A group $(\mathcal{G}, *)$ is a set $\mathcal{G}$ along with a binary operation $*$ such that the following three conditions are met:

1. Associativity: $(a*b)*c = a*(b*c)$
2. Identity: $\exists e \in \mathcal{G}$ s.t. $\forall a \in \mathcal{G}, ae = ea = a$
3. Inverses: $\forall a \in \mathcal{G}, \exists b \in \mathcal{G}$ s.t. $ab = ba = e$

$$\boldsymbol{H}(\boldsymbol{A}) := \begin{bmatrix} \mathbf{0}_{n\times n} & \boldsymbol{A}^{*} \\ \boldsymbol{A} & \mathbf{0}_{m\times m} \end{bmatrix} \in \mathbb{C}^{(m+n)\times(m+n)}$$

A ring $(\mathcal{R}, *_1, *_2)$ with $*_2$ that commutes such that

$$a *_2 b = b *_2 a \ \forall a, b \in \mathcal{R}$$

A ring $(\mathcal{R}, *_1, *_2)$ is a set $\mathcal{R}$ together with 2 binary operations $*_1$ and $*_2$ such that

1. $(\mathcal{R}, *_1)$ is an abelian group with identity
2. $*_2$ is associative:
   $\forall a, b, c \in \mathcal{R}, (a *_2 b) *_2 c = a *_2 (b *_2 c)$
3. $*_2$ distributes over $*_1$ from left and right:
   $a *_2 (b *_1 c) = (a *_2 b) *_1 (a *_2 c)$
   $(a *_1 b) *_2 c = (a *_2 c) *_1 (b *_2 c)$

$$\boldsymbol{R}^{*}\boldsymbol{R}\boldsymbol{x} = \boldsymbol{A}^{*}\boldsymbol{b}$$

A ring $(\mathcal{R}, *_1, *_2)$ with $*_2$ having a identity element.

Normal Matrix	**Rayleigh Quotient**
Implications of A and T in Schur Factorization being Similar	**Determinant of A**
Why are eigenvalue problems challenging?	**Similar Matrices**
Defective Eigenvalue	**Helpful summary of eigenvalue-revealing factorizations**
Theorem 24.1: What makes λ an eigenvalue	**Theorem 24.2: Algebraic Multiplicity**

The Rayleigh quotient of a vector $\boldsymbol{x} \in \mathbb{R}^m$ is the scalar

$$r(x) = \frac{\boldsymbol{x}^\top \boldsymbol{A} \boldsymbol{x}}{\boldsymbol{x}^\top \boldsymbol{x}}$$

A matrix is normal if $\boldsymbol{A}^* \boldsymbol{A} = \boldsymbol{A} \boldsymbol{A}^*$

$\det \boldsymbol{A} = \prod_{i=1}^{n} \lambda_i$

$\boldsymbol{A}$ and $\boldsymbol{T}$ have the same eigenvalues

Two matrices $\boldsymbol{A}$ and $\boldsymbol{B}$ are similar if there exists a nonsingular $X \in \mathbb{C}^{m \times m}$ such that

$$\boldsymbol{B} = \boldsymbol{X}^{-1} \boldsymbol{A} \boldsymbol{X}$$

1. $\boldsymbol{A}\boldsymbol{x} = \lambda \boldsymbol{x}$: two unknowns $\boldsymbol{x}$ and λ
2. λ is an eigenvalue of $\boldsymbol{A}$ iff

   $\det(\lambda \boldsymbol{I} - \boldsymbol{A}) = 0$

   This is a nonlinear function
3. No finite exact algorithm

- A **diagonalization** $\boldsymbol{A} = \boldsymbol{X} \boldsymbol{\Lambda} \boldsymbol{X}^{-1}$ exists iff $\boldsymbol{A}$ is nondefective
- A **unitary diagonalization** $\boldsymbol{A} = \boldsymbol{Q} \boldsymbol{\Lambda} \boldsymbol{Q}^*$ exists iff $\boldsymbol{A}$ is normal
- A **unitary triangularization** (Schur factorization) $\boldsymbol{A} = \boldsymbol{Q} \boldsymbol{T} \boldsymbol{Q}^*$ ALWAYS exists

- An eigenvalue whose algebraic multiplicity exceeds its geometric multiplicity is defective.
- A matrix is defective if it has at least one defective eigenvalue.

If $\boldsymbol{A} \in \mathbb{C}^{m \times m}$, then $\boldsymbol{A}$ has m eigenvalues, counted with algebraic multiplicity.

In particular, if the roots of p_A are simple, then $\boldsymbol{A}$ has m distinct eigenvalues.

λ is an eigenvalue of $\boldsymbol{A}$ iff

$$p_A(\lambda) = \det(\lambda \boldsymbol{I} - \boldsymbol{A}) = 0$$

Theorem 24.3: Similar Matrices	**Theorem 24.4: Relating algebraic and geometric multiplicity**
Theorem 24.5: Diagonalizable Matrices	**Theorem 24.7: Hermitian Matrix Properties**
Theorem 24.8: What makes a matrix unitarily diagonalizable?	**Theorem 24.9: Schur Factorization**
SVD via A^*A	**QR Algorithim for eigenvalues**
Power Iteration	**Least Squares by Normal Equations**

The algebraic multiplicity of an eigenvalue λ is at least as great as its geometric multiplicity.

If $\boldsymbol{X}$ is nonsingular, then $\boldsymbol{A}$ and $\boldsymbol{X}^{-1}\boldsymbol{A}\boldsymbol{X}$ have the same characteristic polynomial, eigenvalues, and algebraic and geometric multiplicities.

A hermitian matrix is unitarily diagonalizable and its eigenvalues are real.

An $m \times m$ matrix $\boldsymbol{A}$ is nondefective (diagonalizable) iff it has an eigenvalue/spectral decomposition

$$\boldsymbol{A} = \boldsymbol{X}\boldsymbol{\Lambda}\boldsymbol{X}^{-1}$$

EVERY square matrix $\boldsymbol{A}$ has a Schur Factorization

A matrix is unitarily diagonalizable iff it is normal.

1. Take the QR Factorization at each step
   $\boldsymbol{A}_k = \boldsymbol{Q}_k\boldsymbol{R}_k$
2. Let $\boldsymbol{A}_{k+1} = \boldsymbol{R}_k\boldsymbol{Q}_k = \boldsymbol{Q}_k^*\boldsymbol{A}_k\boldsymbol{Q}_k$ which is a unitary similarity transformation so eigenvalues are preserved.
   As $k \to \infty$, $\boldsymbol{A}_k \to$ upper triangular matrix whose diagonal entries converge to $(\lambda_1, ..., \lambda_n)$

1. Form $\boldsymbol{A}^*\boldsymbol{A}$
2. Compute spectral decomposition $\boldsymbol{A}^*\boldsymbol{A} = \boldsymbol{V}\boldsymbol{\Lambda}\boldsymbol{V}^*$
3. Let $\boldsymbol{\Sigma}$ be the nonnegative diagonal square root of Λ
4. Form $\boldsymbol{U}$ with
   $\boldsymbol{u}_i = \frac{\boldsymbol{A}\boldsymbol{v}_i}{\sigma_{ii}}$ for $i = 1, ..., r$ where $r = \text{rank}(\boldsymbol{A})$

1. Form matrix $\boldsymbol{A}^*\boldsymbol{A}$ and vector $\boldsymbol{A}^*\boldsymbol{b}$
2. Compute Cholesky Factorization $\boldsymbol{A}^*\boldsymbol{A} = \boldsymbol{R}^*\boldsymbol{R}$
3. Solve lower-triangular system $\boldsymbol{R}^*\boldsymbol{w} = \boldsymbol{A}^*\boldsymbol{b}$ for $\boldsymbol{w}$
4. Solve the upper-triangular system $\boldsymbol{R}\boldsymbol{x} = \boldsymbol{w}$ for $\boldsymbol{x}$

1. Let $\boldsymbol{x}_{k+1} = \frac{\boldsymbol{A}\boldsymbol{x}_k}{||\boldsymbol{A}\boldsymbol{x}_k||}$
2. Once $\boldsymbol{x}_k$ stabilizes
   $\lambda_1 \approx \frac{\boldsymbol{x}_k^*\boldsymbol{A}\boldsymbol{x}_k}{\boldsymbol{x}_k^*\boldsymbol{x}_k}$

Roots of Unity	**Inverse Roots of Unity**
Inverse Unity Matrix	**Denoising the DFT**
Field	**Vector Space**
Unitary Diagonalization	**Schur Factorization**
Geometric Multiplicity of an Eigenvalue λ	**Characteristic Polynomial**

$\omega^{n-1} = \exp\left\{-\frac{2\pi ik}{n}\right\}$ where $\alpha = \omega^{-1}$

$z_{k,n} = \exp\left\{\frac{2\pi ik}{n}\right\} = \cos\left(\frac{2\pi k}{n}\right) + i\sin\left(\frac{2\pi k}{n}\right)$
For $k = 0, 1, ..., n-1$

$$\boldsymbol{x} \xrightarrow{\boldsymbol{F}_n} \hat{\boldsymbol{x}} \xrightarrow{\text{threshold}} \tilde{\boldsymbol{x}} \xrightarrow{\boldsymbol{F}_n^{-1}} \widehat{\boldsymbol{x}}$$

$$\boldsymbol{F}_n = \begin{bmatrix} (\omega^0)^0 & (\omega^0)^1 & \cdots & (\omega^0)^{n-1} \\ (\omega^1)^0 & (\omega^1)^1 & \cdots & (\omega^1)^{n-1} \\ \vdots & \vdots & \ddots & \vdots \\ (\omega^{n-1})^0 & (\omega^{n-1})^1 & \cdots & (\omega^{n-1})^{n-1} \end{bmatrix}$$

A vector space $(\mathcal{V}, +)$ is an abelian group together with a field $(\mathcal{F}, *_1, *_2)$ along with a function $\mathcal{F} \times \mathcal{V} \to \mathcal{V}$, $(a, \boldsymbol{v}) \mapsto \boldsymbol{u} \in \mathcal{V}$, called scalar multiplication that obeys the following laws for $a, b \in \mathcal{F}$ and $\boldsymbol{u}, \boldsymbol{v} \in \mathcal{V}$:

1. $a(\boldsymbol{u} + \boldsymbol{v}) = a\boldsymbol{u} + a\boldsymbol{v}$
2. $(a + b)\boldsymbol{v} = a\boldsymbol{v} + b\boldsymbol{v}$
3. $a(b\boldsymbol{v}) = (ab)\boldsymbol{v}$

A field $(\mathcal{F}, *_1, *_2)$ is a commutative ring with unity in which every nonzero element is a unit ie where every nonzero element has a $*_2$ inverse

Examples: $\mathbb{Q}, \mathbb{R}, \mathbb{C}$

$\boldsymbol{A} = \boldsymbol{QTQ}^*$ where

- $\boldsymbol{Q}$ is unitary
- $\boldsymbol{T}$ is an upper triangular matrix called the Schur form of $\boldsymbol{A}$

If $\boldsymbol{A} \in \mathbb{R}^{m\times m}$ has m linearly independent, orthogonal eigenvectors then there exists a unitary matrix $\boldsymbol{Q}$ such that

$$\boldsymbol{A} = \boldsymbol{Q\Lambda Q}^*$$

$p_A(z) = \det(z\boldsymbol{I} - \boldsymbol{A})$

The maximum number of linearly independent eigenvectors that can be found, all with the same eigenvalue λ

Algebraic Multiplicity of an Eigenvalue λ	**Similarity Transformation**
Diagonalizable	**Graph** $G = (V, E)$
Directed Graph (digraph)	**Adjacency Matrix**
Walk	**Trail**
Path	**Cycle**

Let $X \in \mathbb{C}^{m\times m}$ be nonsingular. Then

$$\boldsymbol{A} \mapsto \boldsymbol{X}^{-1}\boldsymbol{A}\boldsymbol{X}$$

The multiplicity of the root of p_A where:

$$p_A(z) = (z-\lambda_1)(z-\lambda_2)\dots(z-\lambda_n)$$

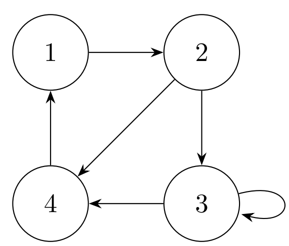
ie if $\lambda_1 = \lambda_2$, the algebraic multiplicity of that eigenvalue is 2

Consists of a finite set of vectors V and a set of edges E

A matrix is diagonalizable if it is nondefective

Reference digraph on its flashcard

$$\boldsymbol{A} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 0 \end{bmatrix}$$



A walk in which no edge is repeated (but vertices may repeat)

A walk of length k from vertex j to vertex i is a sequence of k edges $j \to v_1 \to \dots \to v_k \to i$.

Vertices and edges may both repeat

A closed trail whose interior vertices form a path

A walk in which no vertex is repeated

NOTE	**NOTE**
Nonnegative Matrix	**Stochastic Matrix**
Stochastic Process	**Irreducibility of a Stochastic Matrix $\boldsymbol{P}$**
NOTE	**NOTE**
Monomial Basis	**Norm**

A self loop guarantees **aperiodicity** of a digraph	From most to least general path $\subset$ trail $\subset$ walk
A nonnegative matrix whose rows each sum to 1. ie $\sum_{j=1}^{m} a_{ij} = 1$	$\boldsymbol{A} \in \mathbb{R}^{m \times m}$ such that $a_{i,j} \geq 0 \ \forall i, j$
$\boldsymbol{P}$ is irreducible if for every pair of indices i, j, there exists some $n \geq 1$ such that $(\boldsymbol{P}^n)_{ij} > 0$	A collection of random variables $\{X_t\}_{t \in \mathcal{T}}$ defined on the probability space (Ω, F, P) and indexed by the set $\mathcal{T}$
Know inner and outer meshes (HW 13 2c)	Stochastic matrices are connected to power iteration. Review HW12 3g to see how. Computing the stationary vector of an irreducible stochastic matrix IS power iteration
$\|\|\boldsymbol{x}\|\| = \sqrt{\langle \boldsymbol{x}, \boldsymbol{x} \rangle}$	$f_1(x) = 1, f_2(x)x, f_3(x) = x^2, ..., f_{d+1}(x) = x^d$

Fourier Approximation	**odds**
Companion Matrix	**NOTE**
Strongly connected digraph	**Spectral radius of A**
Stationary Vector of P	**Important Markov Chain Idea**
Circulant Matrices	**NOTE**

$\frac{p}{1-p}$	$f(x) = a_0 + \sum_{k=1}^{\infty} a_k \cos\left(\frac{2\pi}{L}kx\right) + b_k \sin\left(\frac{2\pi}{L}kx\right)$ where • $a_0 = \frac{1}{L}\int_0^L f(x)dx$ • $a_k = \frac{2}{L}\int_0^L \cos\left(\frac{2\pi}{L}kx\right) f(x)dx$ • $b_k = \frac{2}{L}\int_0^L \sin\left(\frac{2\pi}{L}kx\right) f(x)dx$
For companion matrix $\boldsymbol{C}_n$ $\det\left(x\boldsymbol{I} - \boldsymbol{C}_n\right) = g(x)$	$\boldsymbol{C}_n = \begin{bmatrix} 0 & 0 & \cdots & 0 & -a_0/a_n \\ 1 & 0 & \cdots & 0 & -a_1/a_n \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & 1 & -a_{n-1}/a_n \end{bmatrix}$ $g(x) = \frac{a_0}{a_n} + \frac{a_1}{a_n}x + ... + \frac{a_{n-1}}{a_n}x^{n-1} + x^n$
$\rho\left(\boldsymbol{A}\right) = \max_j \lvert\lambda_j\rvert$	A digraph is strongly connected if for every pair of vertices i, j there exists a walk from i to j ie there exists k s.t. $\left(\boldsymbol{A}^k\right)_{ij} > 0$
In predicting values of the future, it does not matter where you have been, only where you are at the current state	A vector $\boldsymbol{\pi} s.t.$ $\boldsymbol{\pi}\boldsymbol{P} = \boldsymbol{\pi}$
A circulant matrix $\boldsymbol{C}$ is determined by its first column The eigenvalues of $\boldsymbol{C}$ are the DFT of its first column $\boldsymbol{C} = \boldsymbol{F}_n^{-1}\mathrm{diag}\left(\boldsymbol{F}_n\boldsymbol{c}\right)\boldsymbol{F}_n$	Completely connected stochastic matrices (Think the tile game)

$\boldsymbol{S}_n$	`runif2()`
Discrete Convolution	**Continuous Convolution**
Linear Independence	**Linear Subspace**
Linear Span	**Basis**
Dimension	**NOTE**

Look at code for this!

A lower triangular matrix of 1s

$f_Y(y) = \int_{\mathbb{R}} f_{X_1}(t) f_{X_2}(y-t) dt$

$f_Y(y) = \sum_{t \in \mathbb{Z}} f_{X_1}(t) f_{X_2}(y-t)$

A linear subspace/vector subspace $\mathcal{W}$ of a vector space $\mathcal{V}$ is a non-empty subset of $\mathcal{V}$ that is closed under vector addition and scalar multiplication.

The elements of a subset $\mathcal{G}$ of a F-vector space $\mathcal{V}$ are linearly independent if no element of $\mathcal{G}$ can be written as a linear combination of the other elements of $\mathcal{G}$.

A subset of a vector space whose elements are linearly independent and span the vector space.

Every vector space has at least one basis.

Given a subset $\mathcal{G}$ of a vector space $\mathcal{V}$, the linear span (span) of $\mathcal{G}$ is the smallest linear subspace of $\mathcal{V}$ that contains $\mathcal{G}$

Also the set of all linear combinations of the elements of $\mathcal{G}$

The 2-norm of a vector and its scaled DFT are the same.

The cardinality shared by all bases of a vector space.

Fast Fourier Transform (FFT)	**`rchaos()` idea**
Fitted values	**Residual Vector**

Set a seed x_1 then $x_n = r(x_{n-1})\,(1 - x_{n-1})$	1. Split $\boldsymbol{x}$ into two vectors of half the lengths, one half is $\boldsymbol{x}_o$ (odds) and the other $\boldsymbol{x}_e$ (evens). 2. Compute the FFT of each, call them $\hat{\boldsymbol{x}}_0$ and $\hat{\boldsymbol{x}}_e$ 3. Add element-wise, the sum of the first vector and the element-wise product of the second with the powers of $\boldsymbol{\omega}$ (same as from DFT). 4. Do the same with difference. Output: $\begin{bmatrix}\hat{\boldsymbol{x}}_o + \boldsymbol{\omega} \odot \hat{\boldsymbol{x}}_e \\ \hat{\boldsymbol{x}}_o - \boldsymbol{\omega} \odot \hat{\boldsymbol{x}}_e\end{bmatrix}$
$(\boldsymbol{I}_n - \boldsymbol{P})\,\boldsymbol{y} = \boldsymbol{y} - \hat{\boldsymbol{y}} = \boldsymbol{e}$ The projection of $\boldsymbol{y}$ onto the orthogonal complement of range($\boldsymbol{X}$) in $\mathbb{R}^n$	$\hat{\boldsymbol{y}} = \boldsymbol{P}\boldsymbol{y}$ response vector projected onto range($\boldsymbol{X}$)